Applications of Network Science

Obesity

Spread of obesity over the social network

Image Credits: Chloe Cushman

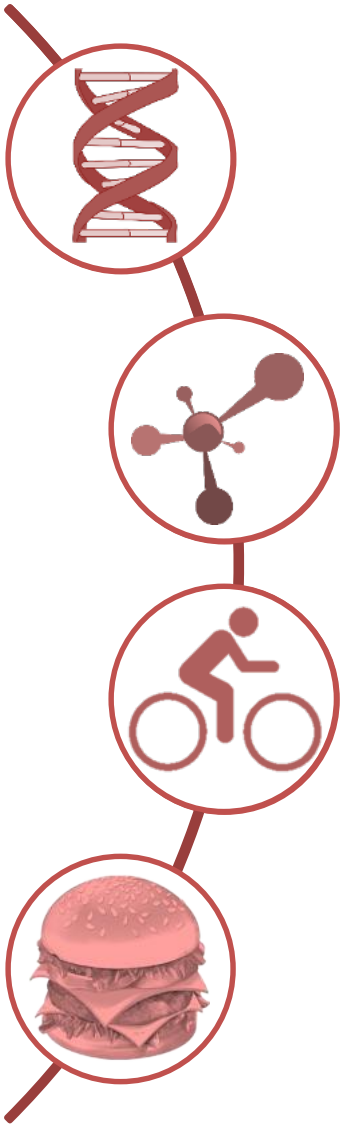
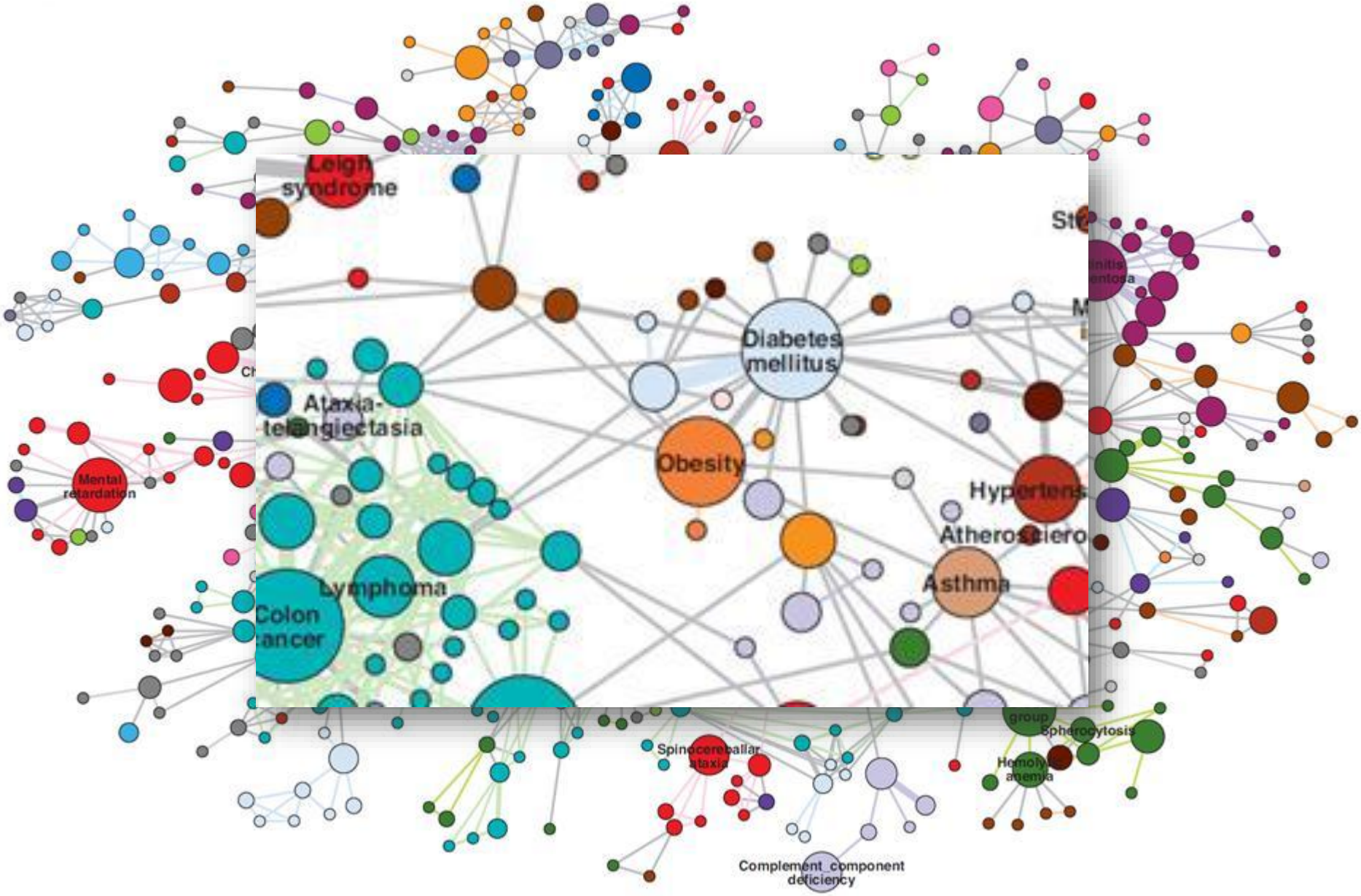


Image Credits: Chloe Cushman



Goeh at al., 104(21), pp. 8685-8690, PNAS (2007).

The NEW ENGLAND JOURNAL of MEDICINE

SPECIAL ARTICLE

The Spread of Obesity in a Large Social Network over 32 Years

Nicholas A. Christakis, M.D., Ph.D., M.P.H., and James H. Fowler, Ph.D.

THE PREVALENCE OF OBESITY HAS INcreased from 23% to 31% over the recent past in the United States, and 66% of adults are overweight.[1,2] Proposed explanations for the obesity epidemic include societal changes that promote both inactivity and food consumption.[3] The fact that the increase in obesity during this period cannot be explained by genetics[4,5] and has occurred among all socioeconomic groups[1] provides support for a broad set of social and environmental explanations. Since diverse phenomena can spread within social networks,[6-10] we conducted a study to determine whether obesity might also spread from person to person, possibly contributing to the epidemic, and if so, how the spread might occur.

Whereas obesity has been stigmatized in the past, attitudes may be changing.[11,12] To the extent that obesity is a product of voluntary choices or behaviors, the fact that people are embedded in social networks and are influenced by the evident appearance and behaviors of those around them suggests that weight gain in one person might influence weight gain in others. Having obese social contacts might change a person's tolerance for being obese or might influence his or her adoption of specific behaviors (e.g., smoking, eating, and exercising). In addition to such strictly social mechanisms, it is plausible that physiological imitation might occur; areas of the brain that correspond to actions such as eating food may be stimulated if these actions are observed in others.[13] Even infectious causes of obesity are conceivable.[14,15]

We evaluated a network of 12,067 people who underwent repeated measurements over a period of 32 years. We examined several aspects of the spread of obesity, including the existence of clusters of obese persons within the network, the association between one person's weight gain and weight gain among his or her social contacts, the dependence of this association on the nature of the social ties (e.g., ties between friends of different kinds, siblings, spouses, and neighbors), and the influence of sex, smoking behavior, and geographic distance between the domiciles of persons in the social network.

Summary

- A quantitative analysis of the nature and extent of the person-to-person spread of obesity as a possible factor contributing to the obesity epidemic.
- Framingham Heart Study
- A densely interconnected social network of 12,067 people assessed repeatedly from 1971 to 2003.
- Body Mass Index = Weight / Height^2 (in meters)
- **Question**: Whether weight gain in one person was associated with weight gain in his or her friends, siblings, spouse, and neighbors?

Glossary

Ego: The person whose behavior is being analyzed.

Alter: A person connected to the ego who may influence the behavior of the ego.

Node: An object that may or may not be connected to other objects in a network. In this study, nodes represent people in the Framingham Heart Study cohorts.

Tie: A connection between two nodes that can be either one-way (directed) or two-way (bilateral). In this study, all family ties (e.g., between siblings and parents) as well as marital ties are bilateral, but friendship ties are directional since a subject may identify someone as a friend who does not necessarily identify that person as a friend in return.

Degree of separation: The social distance between two people as measured by the smallest number of intermediaries between an ego and other members of the network. For a given ego, alters are degree 1, since they are directly connected to the ego. Nodes that are connected to the alters but not to the ego are degree 2 (alters' alters). Nodes that are connected to the alters' alters but not to the ego are degree 3, and so on.

Homophily: The tendency for people to choose relationships with people who have similar attributes.

Induction: The spread of a behavior or trait from one person to another.

Cluster: A group of nodes, each of which is connected to at least one other node in the group.

Connected component: Part of a social network in which all persons have a social tie to at least one other person and no person is connected to a member of any other component of the network.

We considered three explanations for the clustering of obese people. First, egos might choose to associate with like alters ("homophily").[21,23,24] Second, egos and alters might share attributes or jointly experience unobserved contemporaneous events that cause their weight to vary at the same time (confounding). Third, alters might exert social influence or peer effects on egos ("induction"). Distinguishing the interpersonal induction of obesity from homophily requires dynamic, longitudinal network information about the emergence of ties between people ("nodes") in a network and also about the attributes of nodes (i.e., repeated measures of the body-mass index).[25]

We evaluated the role of a possible spread in smoking-cessation behavior as a contributor to the spread of obesity by adding variables for the smoking status of egos and alters at times t and t+1 to the foregoing models. We also analyzed the role of geographic distance between egos and alters by adding such a variable.

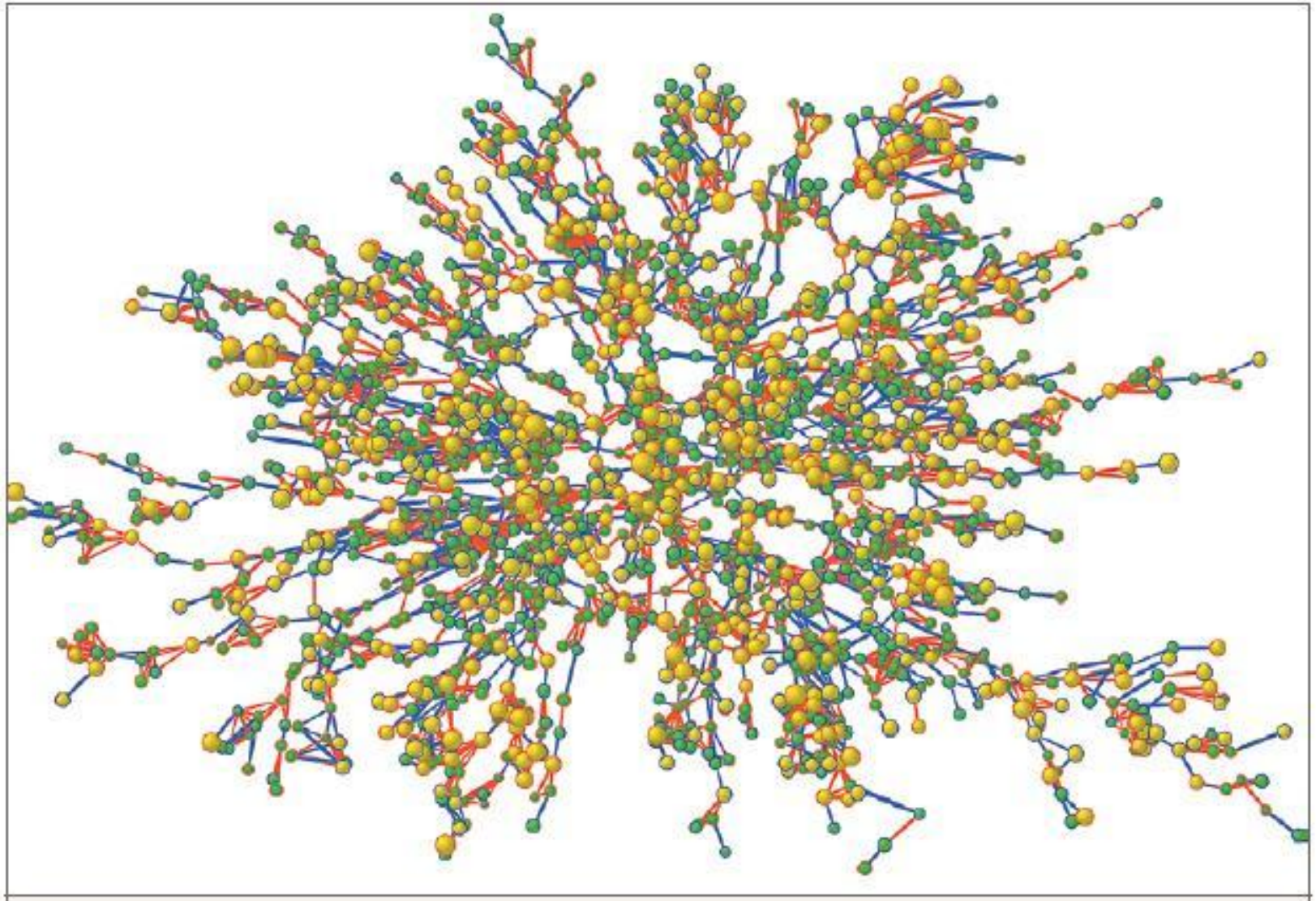
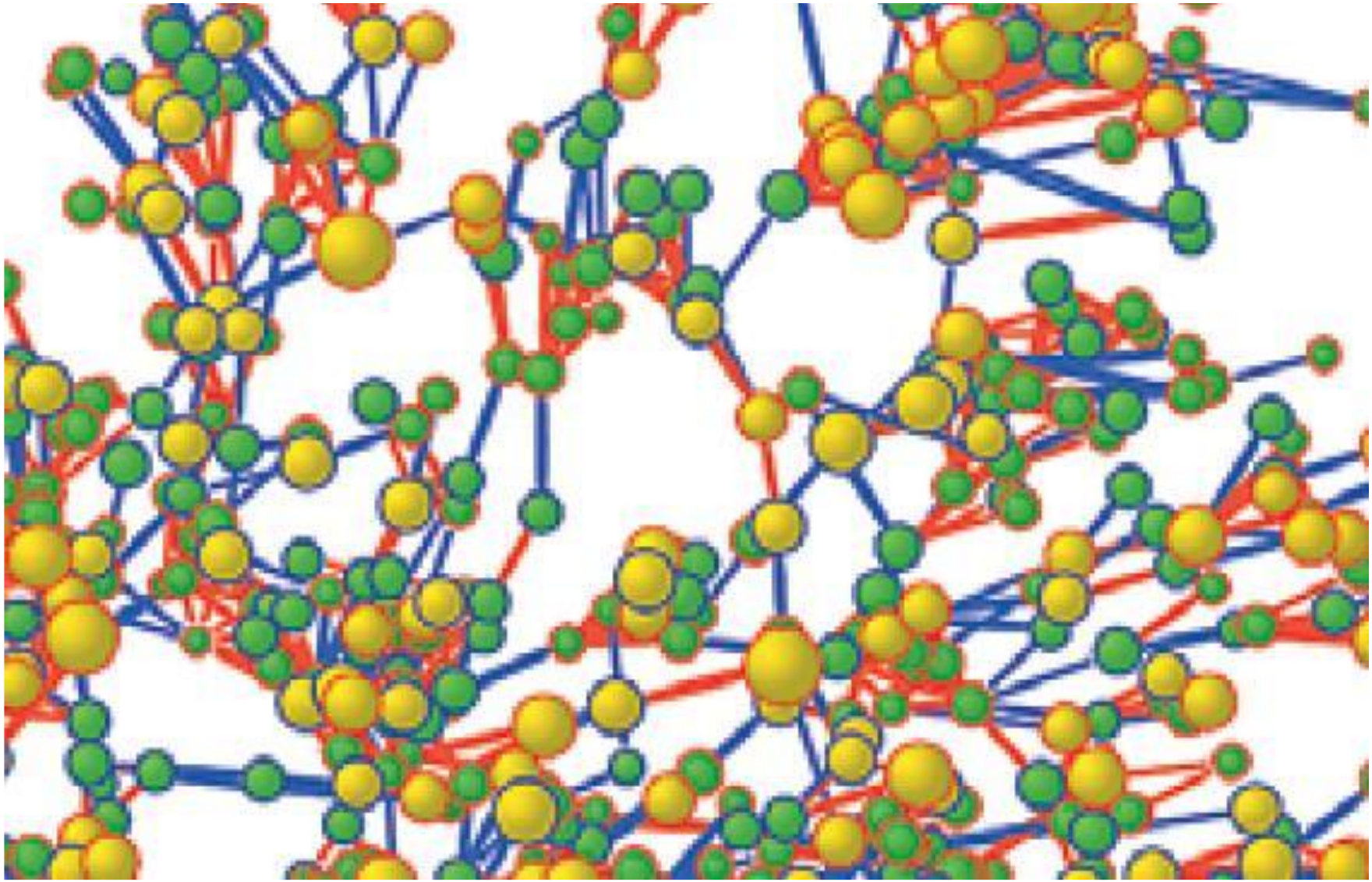
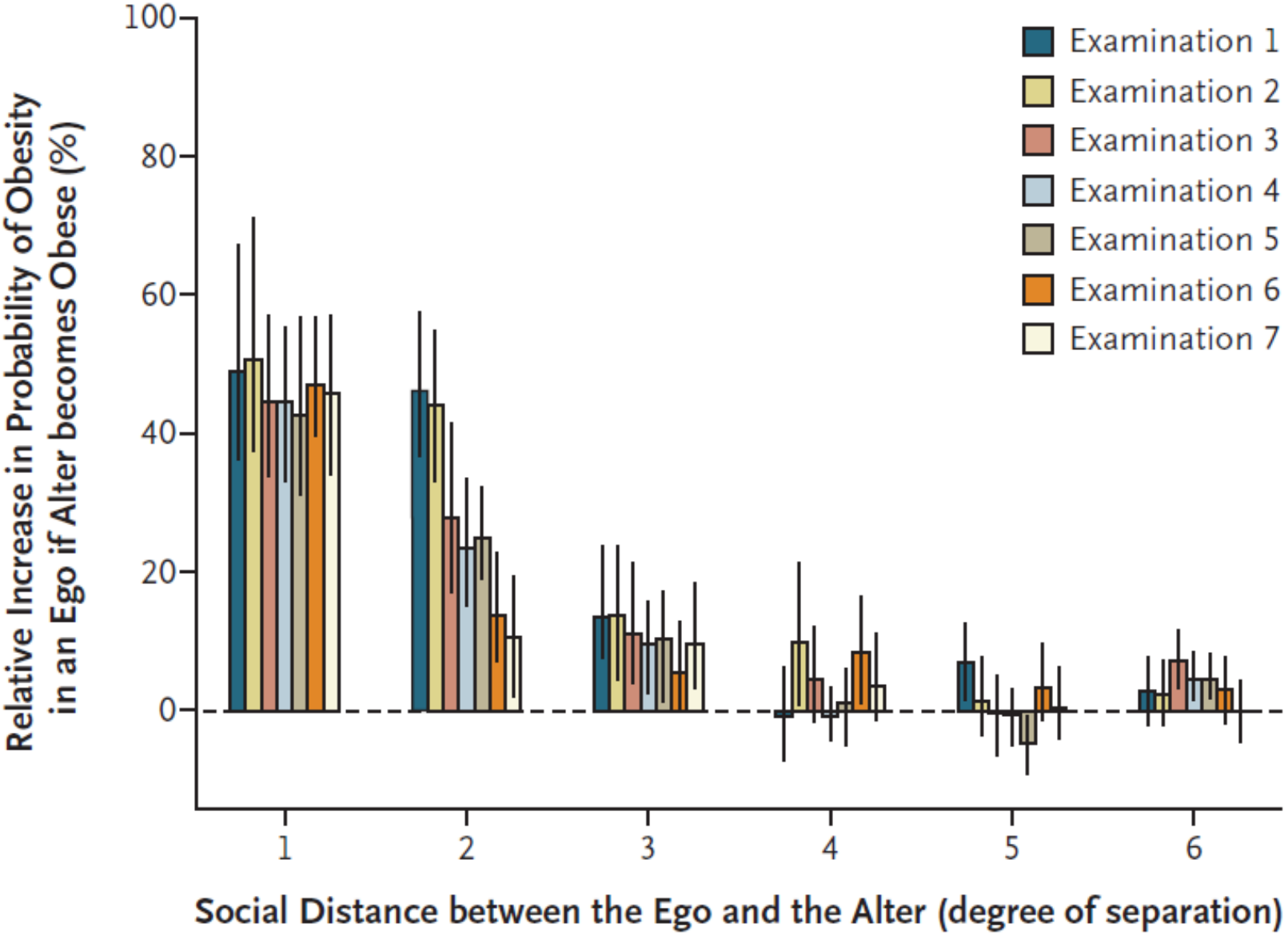


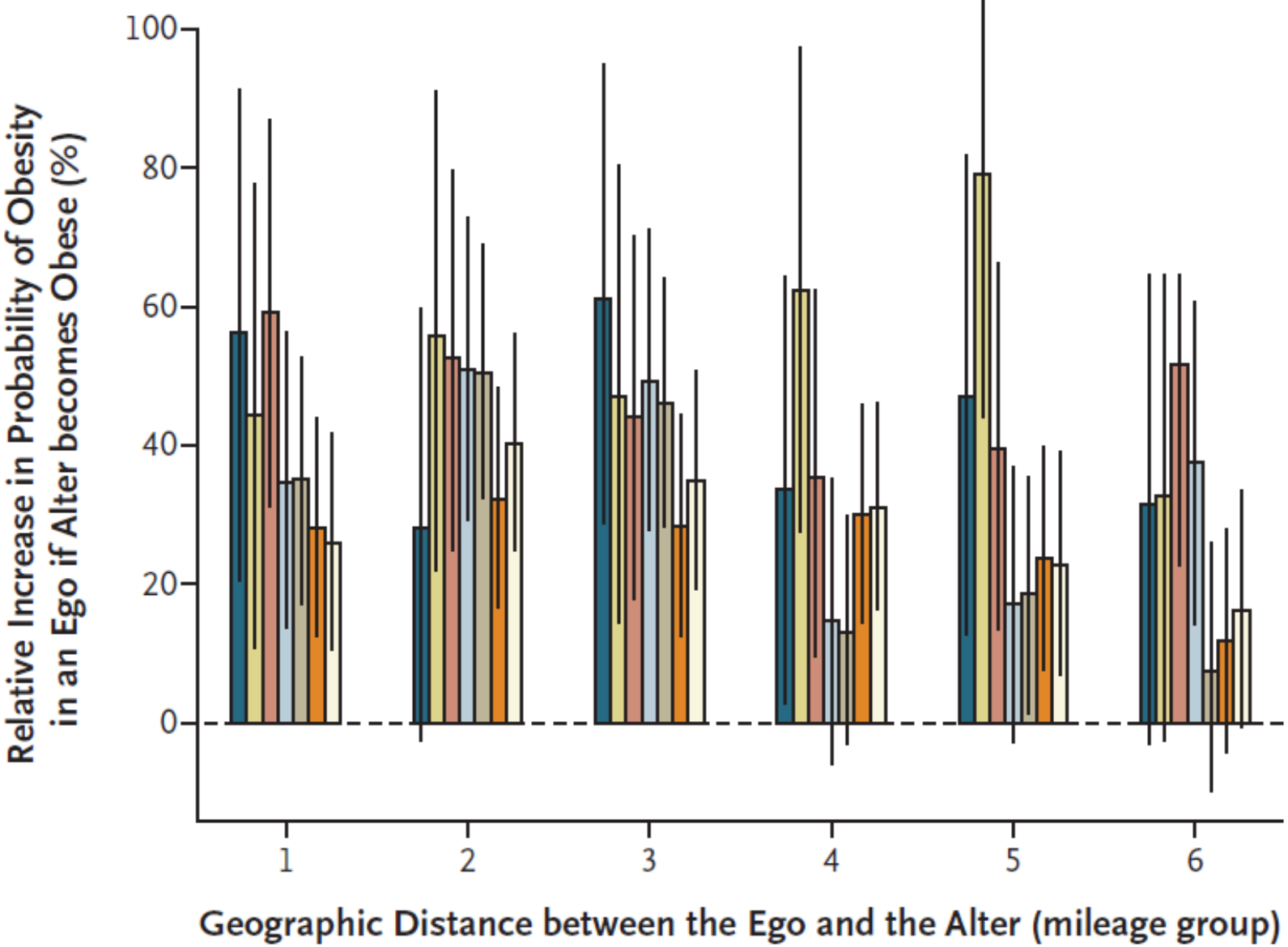
Figure 1. Largest Connected Subcomponent of the Social Network in the Framingham Heart Study in the Year 2000.

Each circle (node) represents one person in the data set. There are 2200 persons in this subcomponent of the social network. Circles with red borders denote women, and circles with blue borders denote men. The size of each circle is proportional to the person's body-mass index. The interior color of the circles indicates the person's obesity status: yellow denotes an obese person (body-mass index, ≥30) and green denotes a nonobese person. The colors of the ties between the nodes indicate the relationship between them: purple denotes a friendship or marital tie and orange denotes a familial tie.

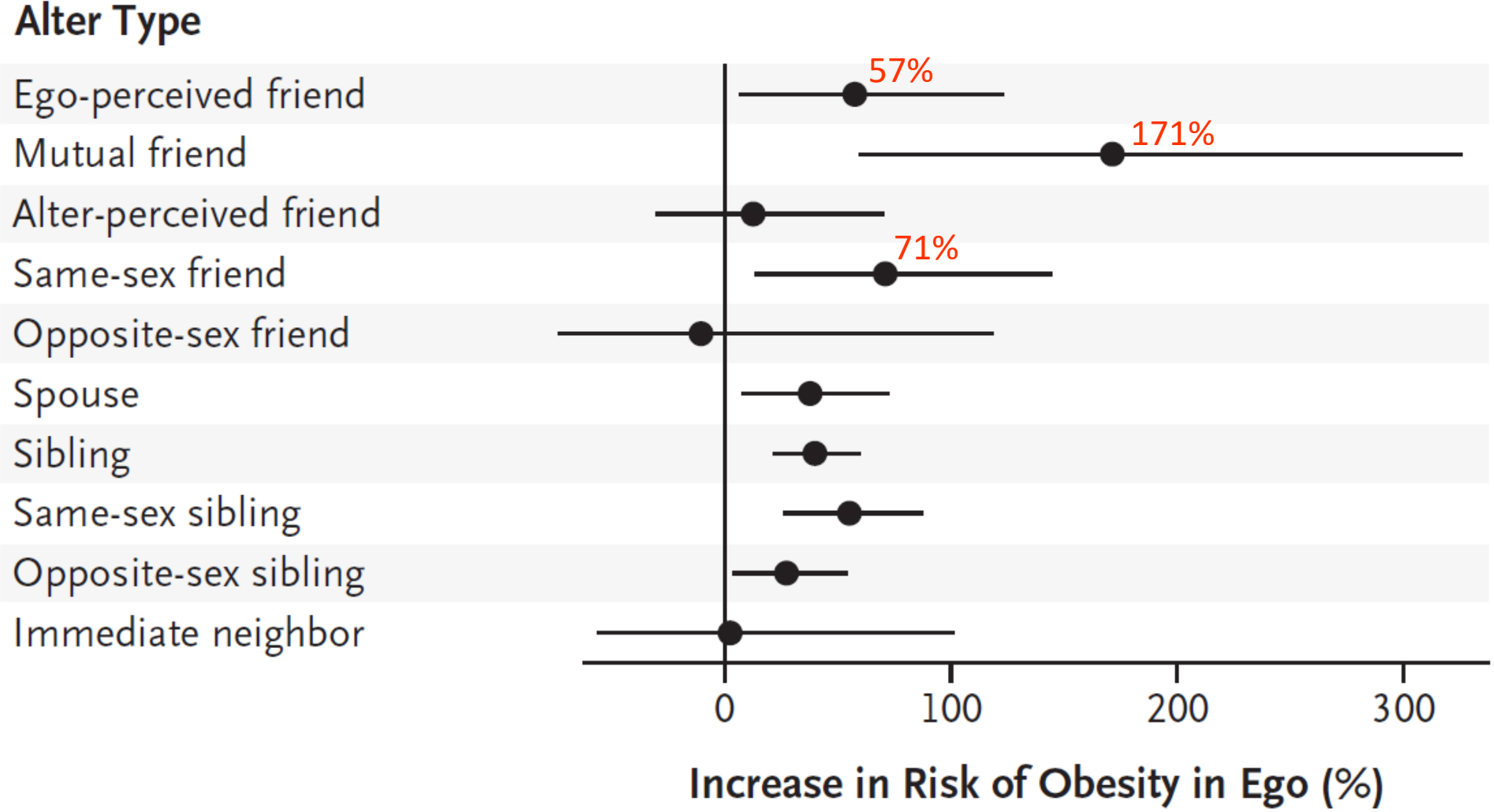




Panel A shows the mean effect of an ego's social proximity to an obese alter; this effect is derived by comparing the conditional probability of obesity in the observed network with the probability of obesity in identical networks (with topology preserved) in which the same number of obese persons is randomly distributed. The social distance between the alter and the ego is represented by degrees of separation (1 denotes one degree of separation from the ego, 2 denotes two degrees of separation from the ego, and so forth). The examination took place at seven time points.



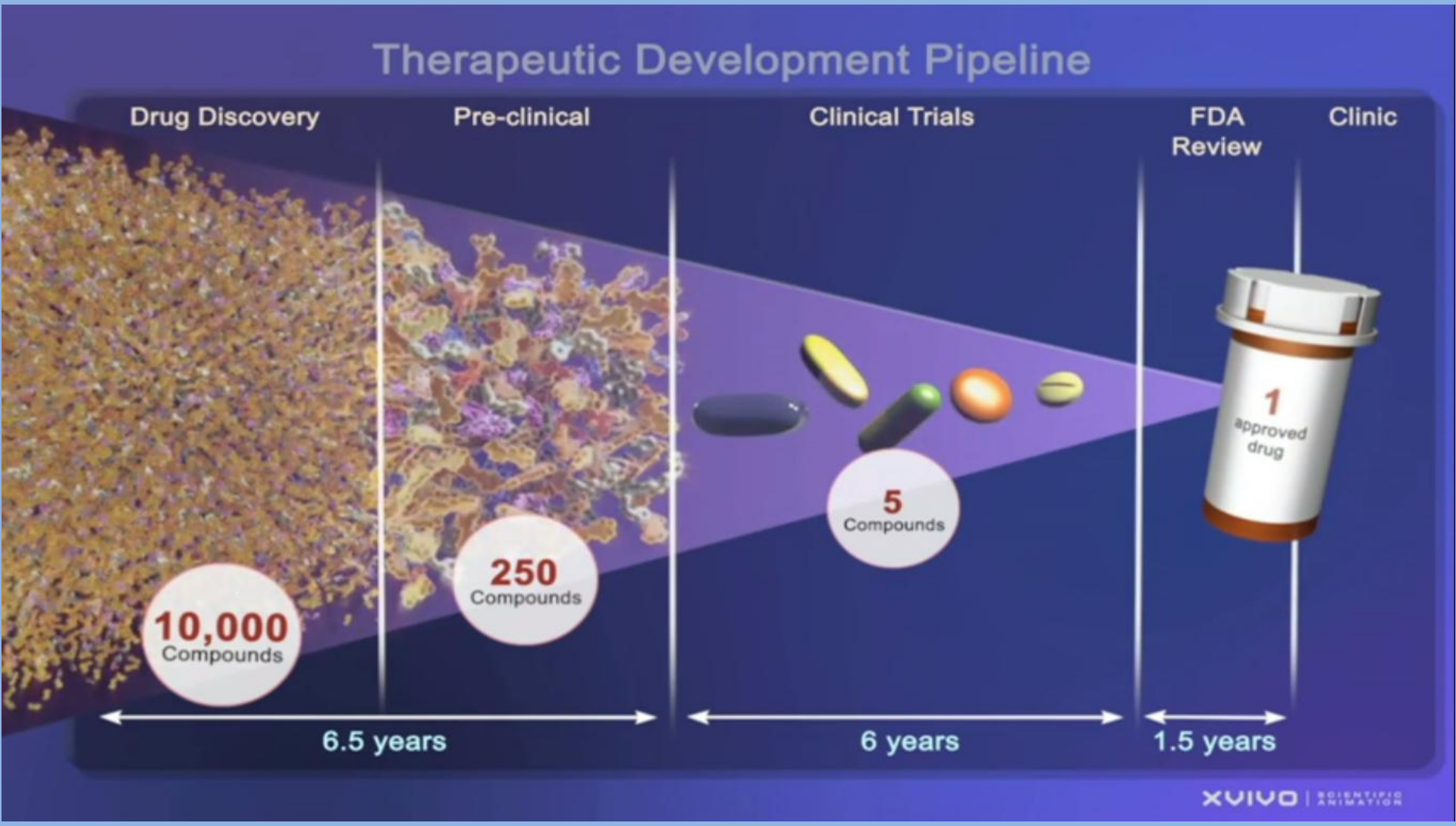
Panel B shows the mean effect of an ego's geographic proximity to an obese alter. We ranked all geographic distances (derived from geocoding) between the homes of directly connected egos and alters (i.e., those pairs at one degree of separation) and created six groups of equal size. This figure shows the effects observed for the six mileage groups (based on their average distance): 1 denotes 0 miles (i.e., closest to the alter's home), 2 denotes 0.26 mile, 3 denotes 1.5 miles, 4 denotes 3.4 miles, 5 denotes 9.3 miles, and 6 denotes 471 miles (i.e., farthest from the alter's home). There is no trend in geographic distance.



Probability That an Ego Will Become Obese According to the Type of Relationship with an Alter Who May Become Obese in Several Subgroups of the Social Network of the Framingham Heart Study.
The closeness of friendship is relevant to the spread of obesity. Persons in closer, mutual friendships have more of an effect on each other than persons in other types of friendships. The dependent variable in each model is the obesity of the ego. Independent variables include a time-lagged measurement of the ego's obesity; the obesity of the alter; a time-lagged measurement of the alter's obesity; the ego's age, sex, and level of education; and indicator variables (fixed effects) for each examination. Mean effect sizes and 95% confidence intervals were calculated by simulating the first difference in the contemporaneous obesity of the alter (changing from 0 to 1) with the use of 1000 randomly drawn sets of estimates from the coefficient covariance matrix and with all other variables held at their mean values.

Conclusions

- Discernible clusters of obese persons with BMI>=30.
- These clusters did not appear to be solely attributable to the selective formation of social ties among obese persons.
- A person's chances of becoming obese increased by 57% if he or she had a friend who became obese in a given interval.
- Among pairs of adult siblings, if one sibling became obese, the chance that the other would become obese increased by 40%.
- If one spouse became obese, the likelihood that the other spouse would become obese increased by 37%.
- These effects were not seen among neighbors in the immediate geographic location.


Therapeutic Development Pipeline
Drug Discovery
Pre-clinical
Clinical Trials
FDA Review
Clinic
10,000 Compounds
250 Compounds
5 Compounds
1 approved drug
6.5 years
6 years
1.5 years
XVIVO | SCIENTIFIC ANIMATION

COMMENTARY Open Access

Drug-target and disease networks: polypharmacology in the post-genomic era

Ali Masoudi-Nejad*, Zaynab Mousavian and Joseph H Bozorgmehr

Abstract

With the growing understanding of complex diseases, the focus of drug discovery has shifted away from the well-accepted "one target, one drug" model, to a new "multi-target, multi-drug" model, aimed at systemically modulating multiple targets. Identification of the interaction between drugs and target proteins plays an important role in genomic drug discovery, in order to discover new drugs or novel targets for existing drugs. Due to the laborious and costly experimental process of drug-target interaction prediction, *in silico* prediction could be an efficient way of providing useful information in supporting experimental interaction data. An important notion that has emerged in post-genomic drug discovery is that the large-scale integration of genomic, proteomic, signaling and metabolomic data can allow us to construct complex networks of the cell that would provide us with a new framework for understanding the molecular basis of physiological or pathophysiological states. An emerging paradigm of polypharmacology in the post-genomic era is that drug, target and disease spaces can be correlated to study the effect of drugs on different spaces and their interrelationships can be exploited for designing drugs or cocktails which can effectively target one or more disease states. The future goal, therefore, is to create a computational platform that integrates genome-scale metabolic pathway, protein–protein interaction networks, gene transcriptional analysis in order to build a comprehensive network for multi-target multi-drug discovery.

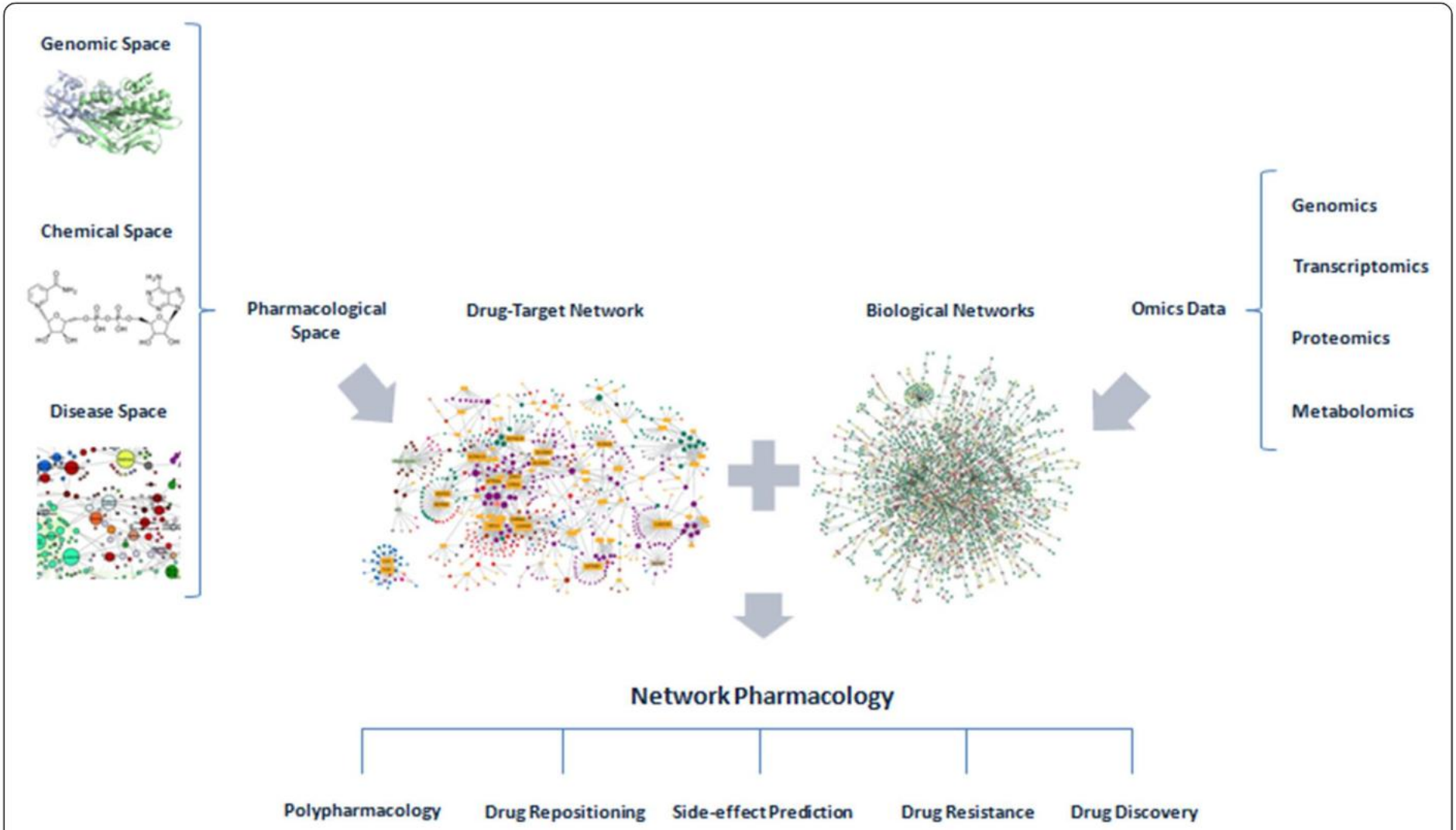
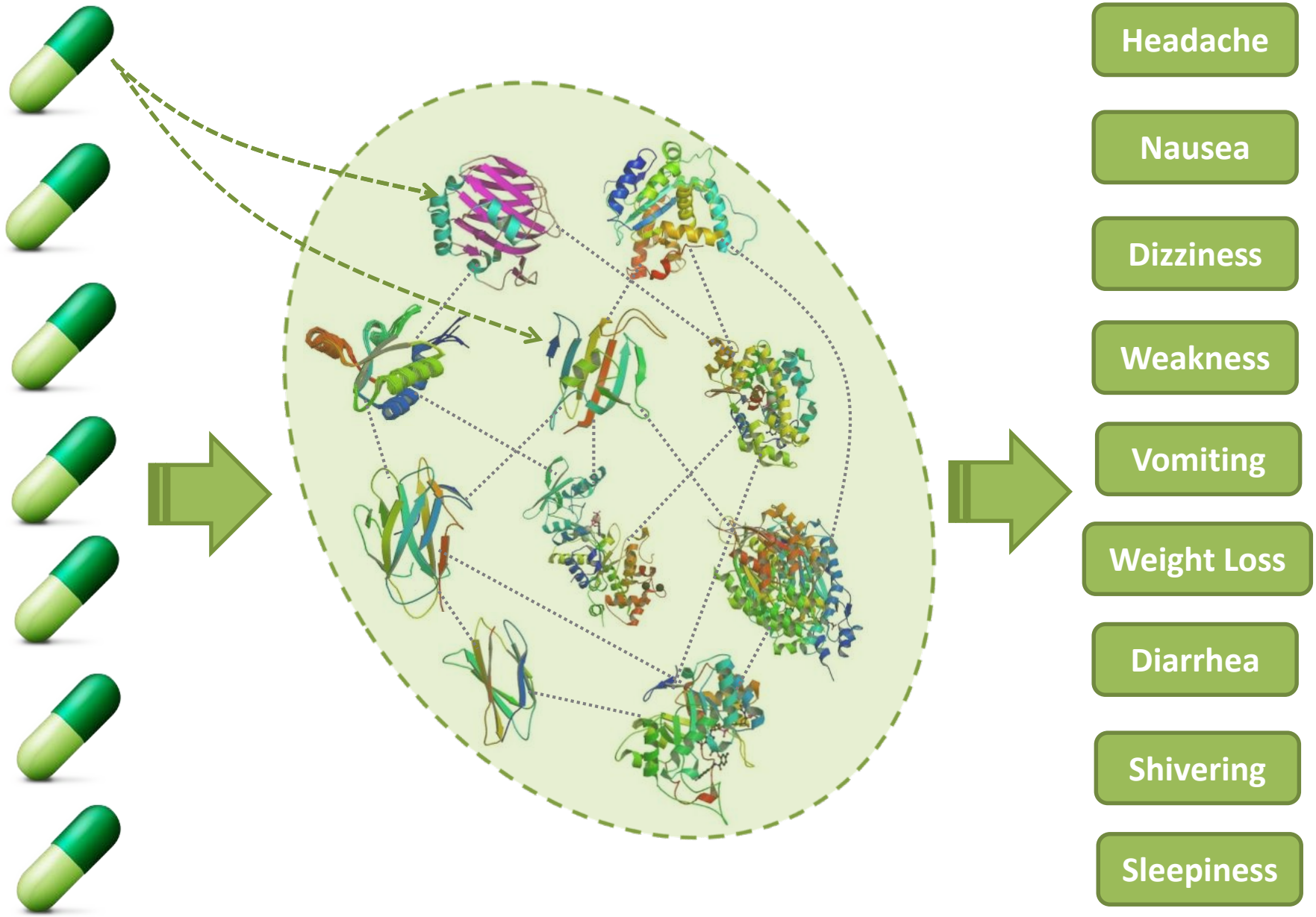


Figure 1 Polypharmacology in the post-genomic era using pharmacological space.

Predicting Adverse Drug Reactions: Machine Learning + Data Analytics



R Kanji, A Sharma and G Bagler*, **Molecular Biosystems** (Royal Society of Chemistry), 11, 2900 (2015);
S Wadhwa, A Gupta, S Dokania, R Kanji and G Bgaler*, **PLoS One** (2018).

Links that speak: The global language network and its association with global fame

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Edited by Kenneth W. Wachter, University of California, Berkeley, CA, and approved November 13, 2014 (received for review June 11, 2014)

Languages vary enormously in global importance because of historical, demographic, political, and technological forces. However, beyond simple measures of population and economic power, there has been no rigorous quantitative way to define the global influence of languages. Here we use the structure of the networks connecting multilingual speakers and translated texts, as expressed in book translations, multiple language editions of Wikipedia, and Twitter, to provide a concept of language importance that goes beyond simple economic or demographic measures. We find that the structure of these three global language networks (GLNs) is centered on English as a global hub and around a handful of intermediate hub languages, which include Spanish, German, French, Russian, Portuguese, and Chinese. We validate the measure of a language's centrality in the three GLNs by showing that it exhibits a strong correlation with two independent measures of the number of famous people born in the countries associated with that language. These results suggest that the position of a language in the GLN contributes to the visibility of its speakers and the global popularity of the cultural content they produce.

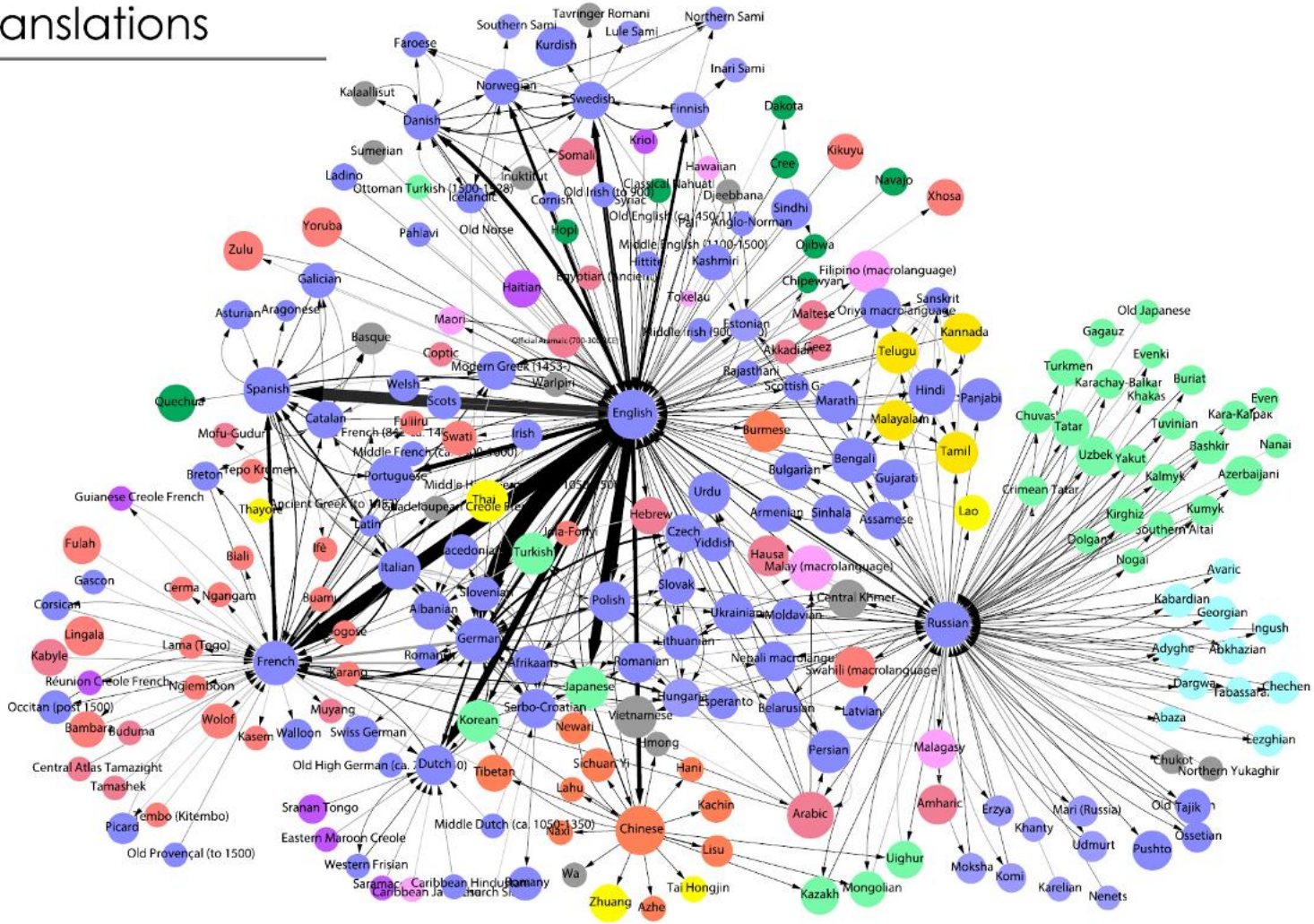
networks | languages | culture | digital humanities | fame

Significance

People have long debated about the global influence of languages. The speculations that fuel this debate, however, rely on measures of language importance—such as income and population—that lack external validation as measures of a language's global influence. Here we introduce a metric of a language's global influence based on its position in the network connecting languages that are co-spoken. We show that the connectivity of a language in this network, after controlling for the number of speakers of a language and their income, remains a strong predictor of a language's influence when validated against two independent measures of the cultural content produced by a language's speakers.

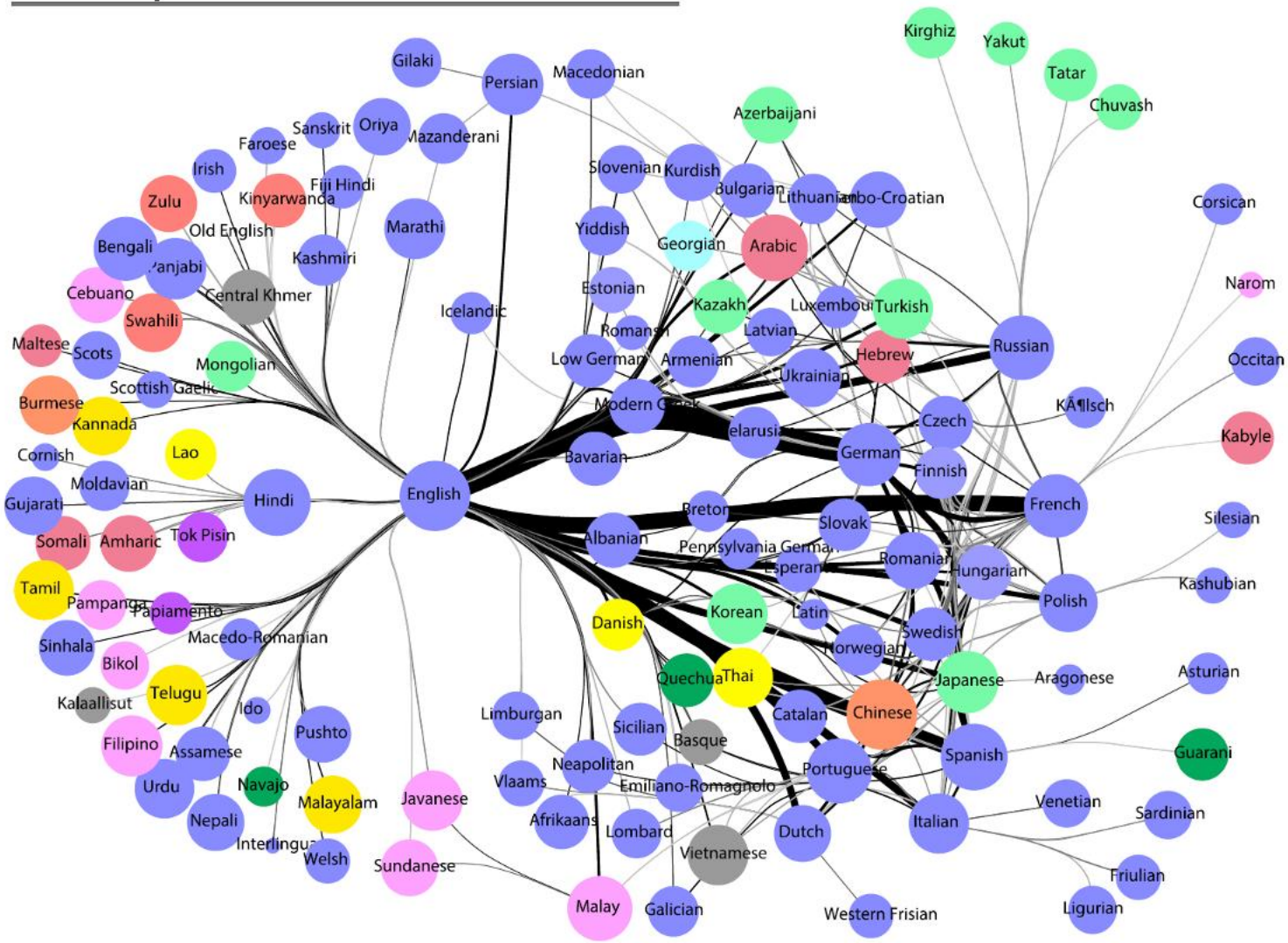
- Of the thousands of languages that have ever been spoken, **only a handful** have become influential enough to be considered **global languages**.
- However, **how do we measure the global influence of a language**?
- **What are the implications of a world in which only a handful of languages are globally influential?**
- In the past, researchers have used a variety of measures to determine the global influence of a language: **the population and wealth of its speakers**.
- While wealth and population approximate a language's influence, **such measures fail to capture the global influence of a language**.
- An alternative method to measure the global influence of a language is to focus on who speaks that language, and in particular, **on how connected the speakers of that language are**.

Book Translations



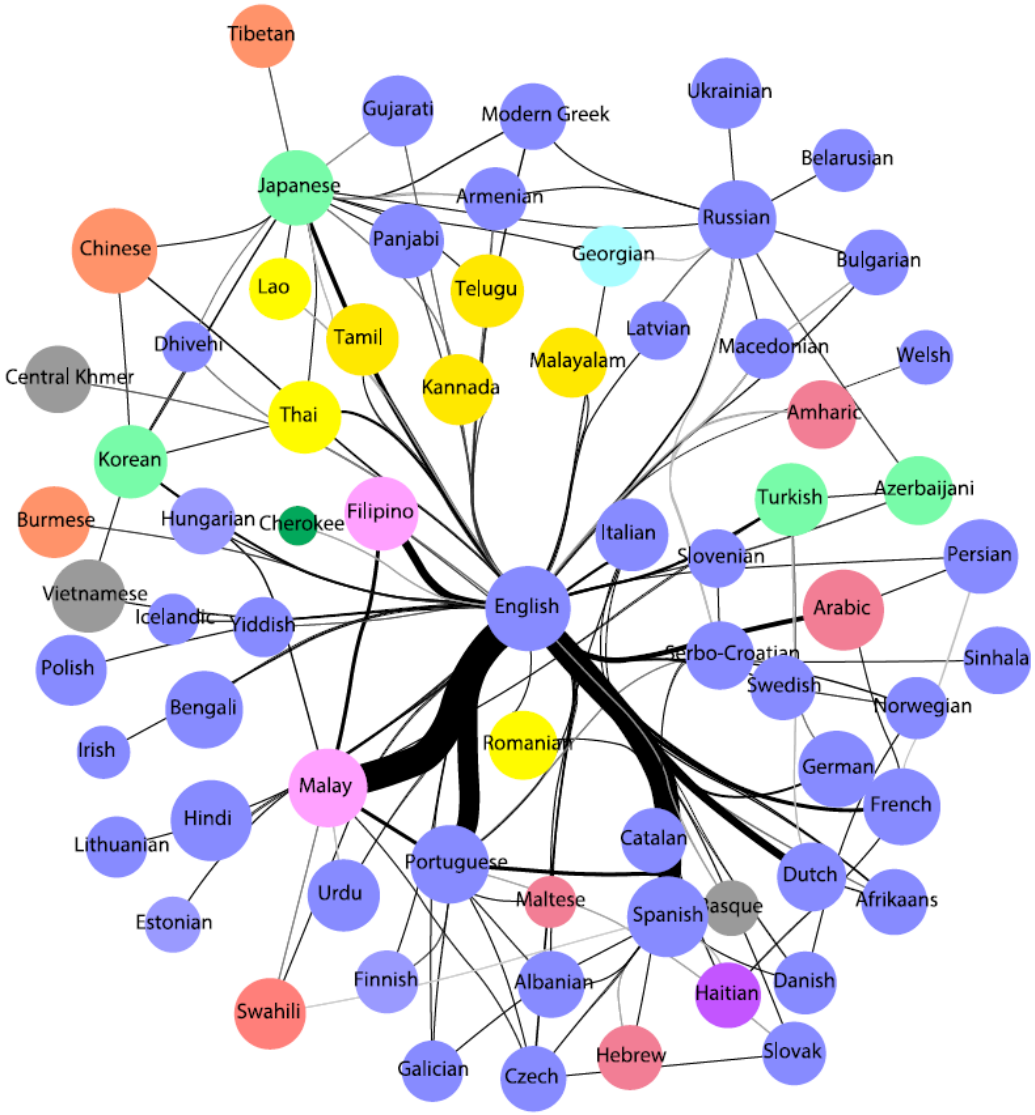
First, we look at a collection of more than 2.2 million book translations compiled by **UNESCO's Index Translationum project**. This dataset allows us to map the network of book translations, which are produced by individuals with a high literary capacity (authors and professional translators) and are shaped by market forces, such as the demand for books in different languages. **Each translation from one language to another forms a connection**.

Wikipedia



Next, we map the network of linguistic coexpressions expressed by **the community of digitally engaged knowledge specialists that edit Wikipedia**. Here, two languages are connected when users that edit an article in one Wikipedia language edition are significantly more likely to also edit an article in another language edition.

Twitter



Finally, we map the network of linguistic coexpressions expressed in **Twitter**. Here, two languages are connected when users that tweet in a language are also significantly more likely to tweet in another language.

Language Family

- Afro-Asiatic
- Caucasian
- Niger-Congo
- Altaic
- Creoles & pidgins
- Other
- Amerindian
- Dravidian
- Sino-Tibetan
- Austronesian
- Indo-European
- Tai
- Uralic

Population

- 1 million
- 10 million
- 100 million
- 1 billion

Link Weight and Color

t-statistic

2.59 — 10

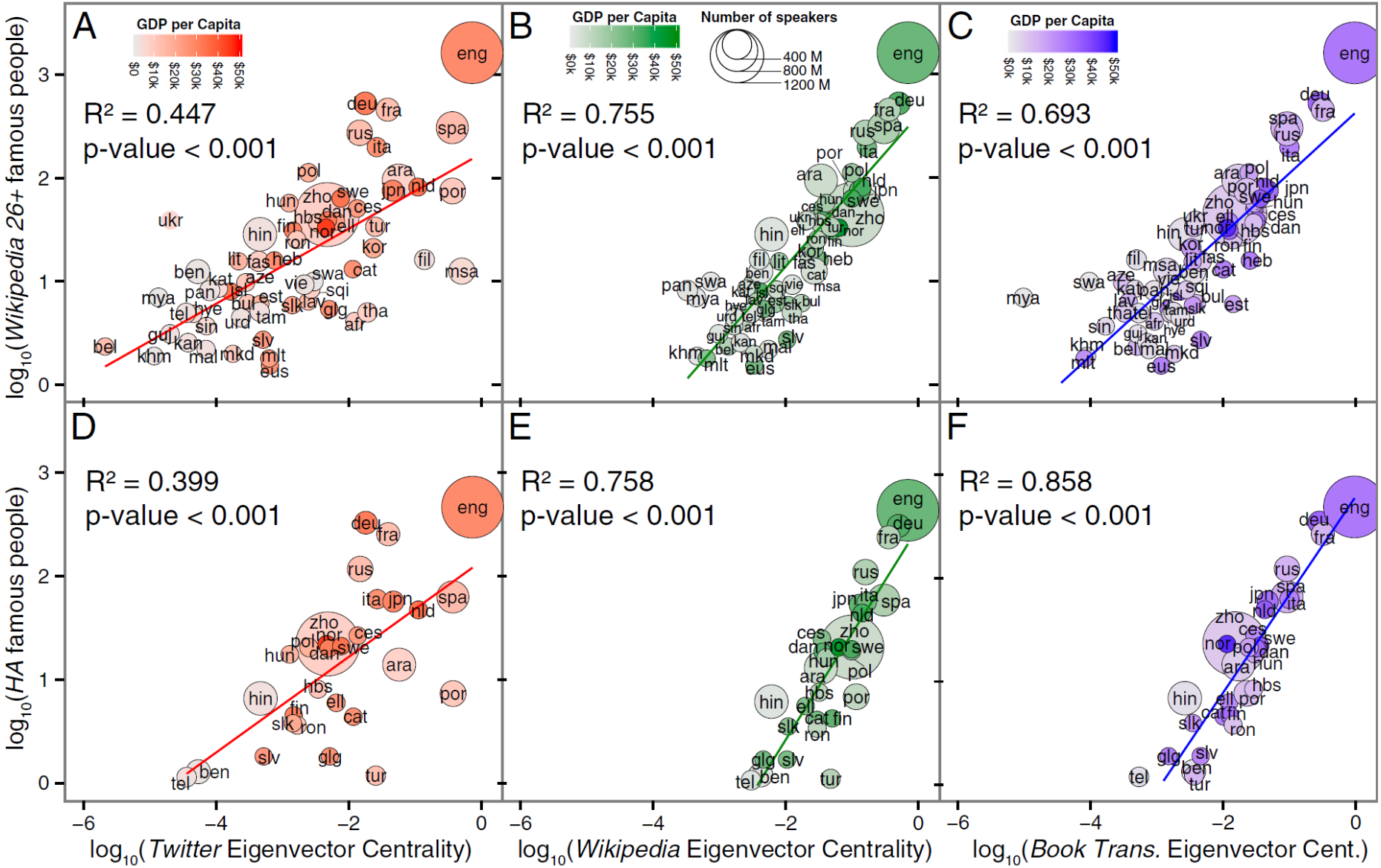
min — co-occurrences (users, editors, translations) — max

6	twitter	994,682
6	wikipedia	49,637
6	book translations	183,329

Measures of global impact of speakers of a language

The scores reported in “Human Accomplishment” (HA).

The number of different language editions in which a biography is present in Wikipedia.



The position of a language in the GLN and the global impact of its speakers. Top row shows the number of people per language (born 1800–1950) with articles in at least 26 Wikipedia language editions as a function of their language's eigenvector centrality in the (A) Twitter GLN, (B) Wikipedia GLN, and (C) book translation GLN. The bottom row shows the number of people per language (born 1800–1950) listed in Human Accomplishment as a function of their language's eigenvector centrality in (D) Twitter GLN, (E) Wikipedia GLN, and (F) book translation GLN. Size represents the number of speakers for each language, and color intensity represents GDP per capita for the language. All subplots report the adjusted R2.